

HOMEWORK 4

1. Prove that a (left) R -module generated by one element is isomorphic to R/I , where I is a (left) ideal in R .
2. Let S be a multiplicative subset of a commutative ring R . For an R -module M define the *localization* $S^{-1}M$ as a module over $S^{-1}R$. (Hint: consider an equivalence relation on $S \times M$.) Show that the correspondence $M \mapsto S^{-1}M$ extends to an exact functor $R\text{-Mod} \longrightarrow S^{-1}R\text{-Mod}$.
3. Let J be a (two-sided) ideal of a ring R and let M be a left R -module. Let JM be the submodule of M generated by the products am for all $a \in J$ and $m \in M$. Show that M/JM has a natural structure of a left R/J -module. Prove that the assignment $M \mapsto M/JM$ extends to a right exact functor $R\text{-Mod} \longrightarrow (R/J)\text{-Mod}$.
4. Let F be a free left R -module with basis $(f_i)_{i \in I}$. Let $J \subset R$ be a (two-sided) ideal of R . Prove that $(f_i + JF)_{i \in I}$ is a basis for the R/J -module M/JM .
5. Let R be a nonzero commutative ring. Prove that if the free R -modules R^n and R^m are isomorphic, then $n = m$. (Hint: Reduce to the case when R is a field.)
6. a) Give an example of a nonzero abelian group A such that $A \oplus A \simeq A$.
b) Let A be a nonzero abelian group A such that $A \oplus A \simeq A$ and $R = \text{End}(A)$. Prove that the free R -modules R^n and R^m are isomorphic for all $n, m > 0$.
7. An abelian group A is called *divisible* if for every $a \in A$ and every nonzero integer n there exists $b \in A$ such that $a = nb$. Prove that an abelian group A is injective (in the category of abelian groups) if and only if A is divisible. (Hint: If A is divisible, in order to extend a group homomorphism $f : X \rightarrow A$ to a group Y containing X , consider the set of all extensions $Y' \rightarrow A$ of f , where Y' is a subgroup of Y containing X and apply Zorn's Lemma.)
8. Prove that the category of abelian groups has enough injectives, i. e., every abelian group is isomorphic to a subgroup of a divisible group. (Hint: Write a group as the quotient of a free group.)
9. Let R be a ring and A an abelian group.
(i) Show that the group $\text{Hom}(R, A)$ has the structure of a left R -module defined by $(rf)(x) = f(xr)$, where $r, x \in R$ and $f \in \text{Hom}(R, A)$.

(ii) Prove that for every left R -module M there is a group isomorphism

$$\mathrm{Hom}_R(M, \mathrm{Hom}(R, A)) \simeq \mathrm{Hom}(M, A)$$

that is natural in M .

(iii) Prove that if A is an injective abelian group, then $\mathrm{Hom}(R, A)$ is an injective left R -module.

10. Prove that for every ring R the category $R\text{-Mod}$ has enough injectives.